AI Career Coach

Existing Solutions, Competitor Comparison & Hackathon Differentiation

Executive finding: Similar products and open-source projects already exist. GitHub code analytics, skill-gap analysis and career coaching are not novel by themselves. A particularly compelling implementation opportunity is a reliable live loop: job requirement - cited repository evidence - gap assessment - real commit - automatically revised assessment. Public feature descriptions alone do not establish whether a competitor lacks a capability.

Scope: This report covers existing solutions and your product opportunity. The original hackathon problem statement has deliberately been omitted.

1. Existing solutions

1. GitRoll - strong commercial competitor

What it does: Analyzes GitHub contributions, developer skills and activity; also serves developer recruiting use cases.

Potential difference: Differentiate with student-specific, job-linked, source-file-cited reassessments after commits.

Reference: https://gitroll.io/

2. SkillProof - close open-source competitor

What it does: Documents resume and GitHub evidence analysis, skill validation, job matching and evidence-backed gaps.

Potential difference: Make webhook-driven, incremental reassessment and a live before/after demo central to your product.

Reference: https://github.com/aryancodex-19/T36-SkillProof

3. SkillBridge - career gap analysis

What it does: Compares candidate profiles or resumes with job descriptions and suggests learning paths.

Potential difference: Demonstrate skills using real checked-in code rather than treating resume claims as proof.

Reference: https://github.com/prithvifr/SkillBridge

4. Skill Gap Analysis - job-market matching

What it does: Uses job-posting APIs and candidate-profile skill comparison to identify gaps.

Potential difference: Connect job requirements to GitHub file-level evidence and commit-triggered updates.

Reference: https://github.com/skill-gap-dev/skill_gap_analysis

5. AI Career Coach - full-stack open-source project

What it does: Describes resume analysis, skill gap detection, job matching, interview preparation and career guidance.

Potential difference: Prioritize reproducible evidence with references to job postings, GitHub file paths and commit hashes.

Reference: https://github.com/DhanshreeD866/AI-Career-Coach

2. Feature comparison

Capability	Existing landscape	Your implementation target
AI career guidance	Widely available	Not a unique headline
Real job requirements	Available in some systems	Retain authentic URL + requirement text
GitHub code analysis	Available in GitRoll, SkillProof	Cite exact paths and source lines
Evidence-based gaps	Available in some projects	Explain positive, partial and missing evidence
Automatic updates on commits	Not established across surveyed projects	Webhook/poll + targeted reassessment
Skill history after commits	Not established across surveyed projects	Before/after skill timeline
Improvement suggestions	Common feature	Tie tasks to missing job evidence

3. Recommended concept: Live Skill Evidence Engine

Instead of only showing a generic "SQL skill gap", the system identifies whether SQL is demonstrated in the analyzed repositories and explains the exact scope of that conclusion. After a commit adds genuine SQL queries, the system reassesses automatically and links the new evaluation to the changed files.

1. Discover a real internship posting and save its source URL and required skills.
2. Connect one or more actual GitHub repositories using GitHub APIs.
3. Analyze the code and store evidence by file path, line, commit and skill.
4. Compare evidence with role requirements and label skills as demonstrated, partially demonstrated or not demonstrated in analyzed scope.
5. Receive a GitHub push webhook (or poll for changes) and identify affected files.
6. Reassess the affected skills and show an evidence-backed before/after history.

Illustrative live demo

Job: Python, SQL and REST APIs required. **Initial repository analysis:** Python demonstrated; SQL not demonstrated in selected code; REST API evidence partial. **Commit:** Add a file such as `queries/report.sql` with real queries. **Webhook:** GitHub detects a push and triggers incremental analysis. **New result:** SQL query-writing is now demonstrated, with the exact file and commit linked. This does not automatically prove database design or production SQL competency.

4. Five features worth building

1. Live GitHub Skill Tracker: Detect pushes and revise affected skills automatically.

2. Skill Evidence Passport: Store requirement, status, supporting path/lines, commit SHA, uncertainty and validation method.

3. AI Gap-to-Code Challenge: Generate a small repository task to demonstrate a specific skill and re-check new code.

4. Career Progress Timeline: Display evidence changes and job-readiness changes over successive commits.

5. Job-Specific Portfolio Builder: Recommend practical improvements to the student's existing projects instead of generic courses.

Prioritization: Build the live tracker and evidence passport first. Add the remaining features only after the core loop is tested end to end.

5. Hackathon feasibility and estimated potential

Real-world usefulness: 5/5

Potential for differentiation: 4/5

Feasibility for a focused prototype: 4/5

Demo impact: 5/5

These are subjective planning assessments, not official judging scores or a prediction of winning.

6. What to show judges

- One authentic internship listing with a working source URL.
- One genuinely connected GitHub repository and actual source-code citations.
- A truthful initial gap assessment that distinguishes "not demonstrated" from "does not know".
- A real code change pushed to GitHub during the demo or captured with verifiable evidence.
- A webhook/polling event that triggers reassessment without manually refreshing the report.
- A changed assessment with new file/commit evidence and an explainable audit trail.

Five-minute demo outline: 0:00-0:40 problem and job source; 0:40-1:40 GitHub analysis; 1:40-2:30 evidence-backed gaps; 2:30-3:45 real code change and detection; 3:45-4:35 updated assessment and audit; 4:35-5:00 innovation and conclusion. Prepare for three minutes of Q&A on evidence accuracy, GitHub security, false positives and scaling.

7. Technical implementation notes

Suggested architecture: React/Next.js frontend; FastAPI or Node.js backend; GitHub REST/GraphQL API plus verified push webhooks; relational database for requirements/evidence/history; code parsing, heuristics and optionally LLM explanations; queue worker for asynchronous repository processing.

Credibility safeguards: Verify webhook signatures; analyze an allowlisted file scope; exclude secrets and sensitive files; keep job URLs and retrieval dates; preserve commit SHA; distinguish observed syntax from production proficiency; never infer lack of ability from missing repository evidence.

8. Conclusion

Recommendation: Proceed with this idea if the team can deliver the working change-detection loop. SkillProof is especially worth studying because it closely overlaps with the evidence-backed skill verification concept. Win differentiation through rigorous citations, automatic updates, trustworthy scope language and a polished live demo,

rather than adding many disconnected AI features.

References

1. GitRoll - strong commercial competitor: https://gitroll.io/

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3. SkillBridge - career gap analysis: https://github.com/prithvifr/SkillBridge

4. Skill Gap Analysis - job-market matching: https://github.com/skill-gap-dev/skill_gap_analysis

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GitHub webhook documentation: GitHub push event

Research note: Feature summaries are based on publicly described capabilities and have not been exhaustively verified by hands-on testing. A feature not found in documentation is not proof it does not exist.